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Procedural Generation of Game Maps as a Constraint Satisfaction Problem: An Euler-based
Conditional Flow Matching Approach

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Flow Matching Approach**

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1 INTRODUCTION

Video games have grown into a multi-billion dollar industry, resulting in the birth of gigantic and complex virtual worlds. One of the core technologies that enable this expansion is Procedural Content Generation (PCG), which consists of computational methods for creating game content algorithmically instead of manually. PCG is essential for generating vast environments, ensuring replayability, and reducing the immense cost and time of manual asset creation (HENDRIKX *et al.*, 2013). Among the most popular applications of PCG in this context is the generation of maps, being present in genres ranging from classic open-world survival games like Minecraft, to Sci-Fi games like No Man’s Sky.

The procedural generation of a valid game map can be formally defined as a Constraint Satisfaction Problem (CSP) (HENDRIKX *et al.*, 2013). A CSP consists of reaching a state where all constraints are satisfied, making it part of the class of NP-Hard problems (HENDRIKX *et al.*, 2013). That means that as the size of the map and the number and complexity of the constraints increase, the time required to find a guaranteed solution can grow exponentially (RUSSELL, NORVIG, 2020). In the context of 2D tile-based map generation, these constraints operate on two levels:

- Local Constraints: Rules concerning the adjacency of individual tiles. For example, a water tile must be adjacent to a sand or grass tile, but never to a lava tile;
- Global Constraints: Rules that ensure the overall coherence of the map, such as the formation of distinct biomes (forests, deserts, oceans), the connectivity of paths, or the logical placement of key locations.

In face of the possibility of the general CSP becoming computationally impossible, this work finds motivation to search for smarter algorithms. Traditional approaches to solving this CSP, such as the Wave Function Collapse (WFC) algorithm, have proven effective but often rely on explicitly defined rulesets derived from example maps (GUMIN, 2016). While powerful, these methods’ need to strictly follow the rulesets hinders their ability to capture bigger-picture design ideas or subtle structural patterns.

In recent years, Deep Generative Models have emerged as a ground-breaking technology, due to their capability of learning complex data distributions from examples. These models, such as Diffusion Probabilistic Models (DDPMs) and Generative Adversarial Networks (GANs),

learn to synthesize new data that mimics the training set (GOODFELLOW *et al.*, 2014; HO, JAIN, ABBEEL, 2020). By training these models on a dataset of existing valid maps, they can learn both the local and global constraints implicitly, without the need for manual rule definition.

While DDPMs have shown great promise, they suffer from a significant drawback: a slow and computationally expensive sampling process (generating a new map, starting from random noise and denoising it step-by-step) that requires hundreds or thousands of iterative steps (HO, JAIN, ABBEEL, 2020). In contrast, this work turns to a more recent and more efficient alternative: Euler-based Flow Matching (LIPMAN *et al.*, 2023). Flow Matching models learn a vector field (a function that determines the direction to move in each point in space) that defines a direct and deterministic path from a simple noise distribution all the way to a valid map (LIPMAN *et al.*, 2023). This is analogous to learning the "flow" that guides random particles into a coherent structure (LIPMAN *et al.*, 2023). The "Euler-based" approach refers to the use of the Euler method—a numerical technique used to solve ordinary differential equations (ODEs)—to traverse this path, allowing for the generation of high-quality samples in very few steps (CHEN *et al.*, 2023).

Therefore, this work proposes and investigates the application of Euler-based Flow Matching as an efficient approach for solving the game map generation CSP. By leveraging its ability to learn how to perform a direct transformation from noise to valid data, we hypothesize that this approach can generate diverse, coherent, and aesthetically plausible maps while drastically reducing the computational overhead associated with other deep generative models.

1.1 OBJECTIVES

This work’s objectives are divided into general and specific objectives.

1.1.1 General Objective

The general objective of this work is to develop, train, and evaluate an Euler-based Flow Matching model for the procedural generation of coherent and diverse maps, framing the task as an implicit solution to a Constraint Satisfaction Problem.

1.1.2 Specific Objectives

The specific objectives are:

- **O1.** To conduct a comprehensive literature review of Procedural Content Generation (PCG), establishing the evolution from classic constraint-based methods (like WFC) to modern deep generative models (like DDPMs and Flow Matching);
- **O2.** To develop a data engineering pipeline to extract, process and convert 2D maps into an adequate format for training, including the division of maps in chunks and the application of one-hot encoding to represent tile geometry and material.
- **O3.** To implement and train an Euler-based Flow Matching model capable of learning the implicit local and global constraints present in the prepared map dataset and generating novel, coherent maps;
- **O4.** To create a control mechanism for the procedural generation, utilizing conditional vectors to guide the model on creating maps with specific characteristics (e.g. presence of lava, biome type, etc.).
- **O5.** To evaluate the implemented Flow Matching model's performance and quality through a two-part comparison:
 - a) A direct, empirical comparison against the WFC algorithm on metric of generation speed and output diversity.
 - b) A literature-based comparison against Denoising Diffusion Probabilistic Models (DDPMs) to argue for the superior computational efficiency of the Flow Matching approach.

2 RELATED WORK

To situate the contributions of this work, it is essential to first survey the landscape of Procedural Content Generation (PCG) and deep generative models. This chapter provides such a review, charting a clear path from the foundational principles of PCG to the cutting-edge techniques that directly motivate this project. The chapter begins by establishing the review methodology in Section 2.1. It then proceeds with a focused analysis of three pivotal publications that, respectively, define the problem space, justify the adoption of deep learning, and introduce the core Flow Matching technique. The chapter culminates in a comparative analysis of these key approaches, identifying the gaps in the current literature that the method proposed in this project aims to address.

2.1 LITERATURE REVIEW METHODOLOGY

This section outlines the systematic methodology employed to identify, screen, and select the relevant literature for this work. The objective was to build a comprehensive understanding of the state-of-the-art in Procedural Content Generation (PCG), focusing on the transition from classic constraint-based methods to modern deep generative models for game map creation. The process was divided into three stages: identifying information sources, executing keyword-based searches, and applying selection criteria to filter the results.

First, the search was conducted across several major academic databases and search engines known for their extensive coverage in computer science and artificial intelligence. The primary sources included the ACM Digital Library, IEEE Xplore, and Google Scholar. Additionally, ArXiv was consulted for pre-print articles to ensure coverage of the latest, cutting-edge research in the rapidly evolving field of generative models.

The search strategy was executed in a multi-stage process to ensure both breadth and depth. The initial phase focused on identifying foundational survey papers using broad search strings such as “procedural content generation survey” and “deep learning for PCG review”. Once these cornerstone papers were identified, a more targeted search was conducted focusing on specific technique and applications. This involved combining technical keywords like “Wave

Function Collapse”, “Generative Adversarial Network”, “Diffusion Models”, and “Flow Matching” with application-specific terms like “level generation” or “map generation”. Further searches were performed to find alternative constraint-based methods (e.g., “Answer Set Programming for procedural content generation”) and papers on evaluation methodologies (e.g., “evaluating PCG metrics”) to establish a comprehensive basis for comparison.

The initial searches yielded a large number of publications. To refine this set to the most pertinent articles, a two-stage filtering process was applied based on a set of inclusion and exclusion criteria. Papers were included if they were: (a) foundational surveys of the PCG field; (b) directly addressed the generation of 2D tile-based maps or levels; (c) proposed or analyzed deep generative models relevant to the task; or (d) were published in high-impact journals or conferences. A strong preference was given to recent publications (post-2020) for topics related to deep learning to ensure the review was current. Conversely, papers were excluded if their primary focus was on non-map content (e.g. narrative, audio), or if they discussed older techniques not directly relevant for comparison. This structured process resulted in the final selection of the key papers analyzed in the following sections.

2.2 A FOUNDATIONAL SURVEY OF PROCEDURAL CONTENT GENERATION FOR GAMES

The work of Hendrix et al. (2013) stands as a seminal contribution to the field, being the first comprehensive survey to formally structure and define the landscape of Procedural Content Generation for Games (PCG-G). The authors were motivated by the exponentially growing unsustainability of manual content creation, which was becoming a bottleneck in game development due to rising resource costs and scalability challenges (HENDRIKX et al., 2013). The paper’s primary objective was to use the spreading need to come up with an effective way to generate content that satisfied the hungry gamer community in terms of quality and quantity. The way the authors found to do this was to organize PCG-G research into a cohesive framework, establishing a common terminology and clear classification system for the various types of game content and the methods for generating them.

The main contribution of the survey that relates to this work, is a dual-taxonomy approach. First, it proposes a six-layered pyramid to classify game content (Figure 2.1), starting

from the most basic elements (Game Bits) and moving up through Game Space, Game Systems, Game Scenarios, Game Design, and finally Derived Content (HENDRIKX et al., 2013). For this work, the “Game Space” layer is particularly relevant, as it formally defines the environment where the game takes place, including the generation of indoor and outdoor maps.

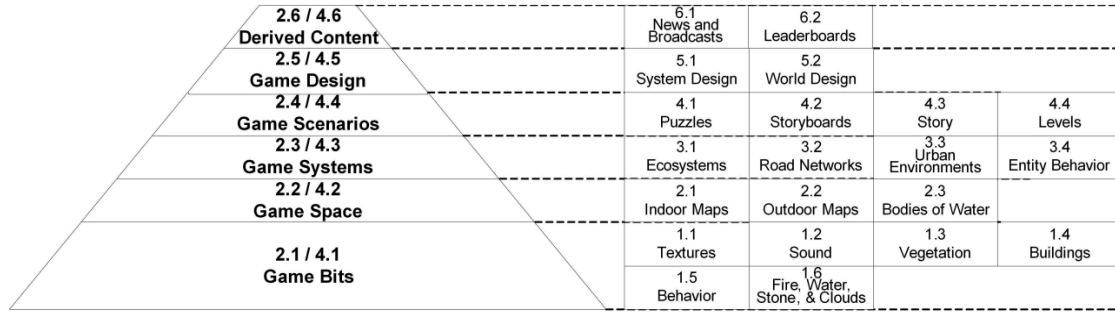


Fig. 2.1. Content hierarchy proposed by Hendrikx et al. (2013).

Second, and most crucially, the paper categorizes generation algorithms, placing “Constraint Satisfaction and Planning” under Artificial Intelligence methods. (HENDRIKX et al., 2013). The authors explicitly define this approach as finding a valid state that satisfies a set of predefined constraints, acknowledging it as an NP-hard problem (HENDRIKX et al., 2013). This classification is fundamental for the present study, as it provides a formal academic validation for framing the procedural generation of maps as a Constraint Satisfaction Problem (CSP), which is the core premise of this project. Furthermore, the survey highlights that while constraint-based methods are precise, they struggle with scalability and complex global structures, which are exactly the limitations that modern deep generative models, as the one proposed in this work, aim to overcome. By outlining the state-of-the-art in 2013, Hendrikx et al. (2013) also implicitly highlights the limitations of classic methods, particularly the reliance on manually defined rules (which behave as the constraints of a CSP) such as what is performed by the WFC algorithm, which will be described in further detail in the subsequent chapters. Therefore, one could interpret the identification of such limitations by Hendrikx et al. (2013) as setting the stage for the subsequent rise of deep learning models capable of learning the CSP constraints implicitly, overcoming the previous manual definition need.

2.3 THE REVOLUTION OF DEEP LEARNING IN PCG

While the foundational survey by Hendrix et al. (2013) structured the field of classic PCG, the subsequent years saw a paradigm shift towards data-driven methods, a transition captured by the survey of Liu et al. (2020) on deep learning for PCG. This work documents the revolution brought by deep generative models, which moved the field from manually defined rulesets to implicitly learned constraints from existing data (LIU et al., 2020). The paper’s primary contribution is to map how different families of deep learning—such as Variational Autoencoders (VAEs) and GANs—were being applied to generate game content, particularly game levels (LIU et al., 2020).

A crucial concept highlighted by Liu et al. (2020) is the challenge of functionality constraints. The authors argue that generating game content is fundamentally different from other generative tasks; a game level must be functional (i.e., playable), and a failure in functionality renders its utility as “essentially zero”. This distinction is instrumental for this work, as it reinforces the necessity of viewing map generation through the lens of a CSP, even when using modern generative models.

The survey further details that early deep learning approaches often treated game levels as simple 2D images (LIU et al., 2020). Models like GANs were trained to replicate the visual and stylistic patterns of existing levels (e.g. given a dataset of multiple valid levels, the model would implicitly learn how to replicate the levels, resulting in ample variety). However the authors emphasize that, unlike natural images, game levels must satisfy strict functionality constraints, such as playability. They review various metrics used to evaluate these models, such as playability rate, visual diversity, they often fail to guarantee functional coherence without complex, domain-specific constraints. This insight directly motivates our investigation into Flow Matching, which offers a more stable training regime and potential for better adherence to global structural constraints compared to the adversarial training of GANs.

2.4 FLOW MATCHING: AN EFFICIENT GENERATIVE MODELING PARADIGM

Building upon the landscape of deep generative models surveyed by Liu et al. (2020), the work of Lipman et al. (2023) introduces Flow Matching (FM), a novel and efficient paradigm

within the broader framework of Continuous Normalizing Flows (CNFs). This approach directly addresses the primary drawback of Denoising Diffusion Probabilistic Models (DPPMs)—their slow and computationally intensive sampling process which requires numerous iterative steps. Flow Matching proposes an alternative by learning a deterministic vector field that defines a direct path from a simple noise distribution to the complex data distribution—which, for this study, would be a valid tile-based 2D map—, thereby enabling significantly faster sample generation.

The core idea behind Flow Matching is to train a neural network to learn the ideal vector field. In this context, a vector field can be described as a massive list of directions, one for each pixel in the map. Meanwhile, a neural network can be understood as a function. It is given two inputs: the current state of the data (which starts as noise and gradually becomes a map) and a “time” value (from 0 to 1, representing how far along the transformation process it currently is). This vector tells *exactly* how to adjust the value (color/tile type) of every single pixel at that specific moment to move it slightly closer to, together with all of the surrounding pixels, looking like a real map. The goal is to learn the vector field that optimally describes the probability path - the smooth, ideal transformation from pure noise at time 0 to valid map data at time 1. In other terms, the neural network should regress the vector field, that is, to learn to accurately predict the ideal directions at every point and time of this transformation.

Lipman et al. (2023) introduced a simulation-free training method called the Conditional Flow Matching (CFM) objective. It does not need to simulate the entire noise-to-map path during each training step, making training much faster. Instead, it learns efficiently by looking at conditional probability paths defined per data sample. This means the network learns from many single examples: for each real map in the training data, it learns the specific directions needed to transform noise into *that* specific map. By learning these individual transformations, the network generalizes to understand the overall transformation process for all maps. Once trained, the model generates new data by solving an Ordinary Differential Equation (ODE) that follows the learned vector field, starting from a random noise sample. The Euler-approach chosen by this work refers to specifically using the simple and fast Euler method to numerically integrate this ODE. The flow the Flow Matching model would follow to generate a valid map step-by-step would be:

1. Start with noise (a grid of random pixel values);

2. Ask the neural network (the vector field): “Based on these current pixel values and time $t = 0$, what small adjustments should I make?”;
3. The network outputs the adjustment vector (directions for every pixel);
4. Apply these tiny adjustments to all pixel values;
5. Slightly increase the time t ;
6. Repeat steps 2-5 many times until $t = 1$. Solving the ODE is this step-by-step adjustment process. The final state of the pixel values at $t = 1$ is the generated map;

Compared to diffusion models, Flow Matching offers several key advantages. Firstly, the CFM objective leads to more stable and often faster training convergence, meaning the neural network learns reliably and reaches its best performance more quickly during the training phase. Secondly, and most critically for this project, the sampling process is significantly more efficient. Because the learned path is direct, high-quality samples can be generated with a considerably lower Number of Function Evaluations (NFE)—the number of times it is needed to ask the neural network for directions—using standard ODE solvers. Thirdly, Flow Matching is not restricted to the probability paths inherent in diffusion processes; Lipman et al. explore the use of paths derived from Optimal Transport (OT), a mathematical theory regarding the most efficient way to morph one shape or distribution into another. These OT paths empirically result in even faster training and sampling due to their straight-line trajectories.

2.5 SPECIALIZED GENERATIVE APPROACHES: GANS, APPLIED DIFFUSION AND CONSISTENCY MODELS

Beyond the foundational surveys and the general generative architectures discussed above, recent literature has focused on applying specific deep learning families to procedural content generation, highlighting both their potential and their limitations relative to the goals of this work.

Silva et al. (2025) provide a comprehensive review of Procedural Game Level Generation using GANs. Their work analyzes the extensive use of GANs to replicate visual patterns in game levels. However they identify critical unresolved challenges, specifically the difficulty GANs face in satisfying strict functional constraints (such as playability and connectivity) without complex post-processing or hybrid search methods. Furthermore, they discuss the issues of

training instability and “mode collapse” (in which the model fails to generate sufficiently diverse outputs), reinforcing the need for more stable generative paradigms, such as the Flow Matching approach this work proposes.

In the domain of diffusion models, Ren (2024) demonstrates the efficacy of Denoising Diffusion Probabilistic Models (DDPMs) in the environment of the game Minecraft. The author successfully trained a DDPM to generate coherent 3D structures with voxels (blocks that are the 3D equivalent to tiles), proving that iterative denoising processes *can* capture the spatial dependencies of voxel-based worlds effectively. However, the work also implicitly highlights the computational cost associated with the diffusion sampling process, which requires a large number of steps to converge to the desired high-quality result. This trade-off between output quality versus computational cost and generation speed serves as a direct motivation for this study to explore more efficient alternatives.

Finally, the most recent evolution in efficient sampling is represented by Consistency Models (Song et al., 2023). While diffusion models rely on iterative noise removal, Consistency Models aim to map any point x_t on the probability trajectory directly to the data origin x_0 in a single step. This is achieved by enforcing a self-consistency property: the model’s prediction of the final output must remain invariant regardless of the time step from which the prediction is made. Formally, for any points on the same trajectory, the model must satisfy $f(x_t, t) = f(x_{t'}, t')$. While this paradigm offers the theoretical possibility of one-step generation, training these models directly can be unstable and complex. Flow Matching, therefore, serves as a robust middle ground: by learning a deterministic vector field with straight trajectories via Optimal Transport, it enables highly efficient sampling with standard ODE solvers without the rigid constraints and instability often associated with enforcing self-consistency.

In light of this comparative analysis, this work adopts the standard Euler-based Flow Matching framework, specifically leveraging the Conditional Flow Matching (CFM) objective. This objective simplifies the training process by regressing a time-dependent vector field that defines optimal paths between noise and data samples, avoiding computationally expensive simulations of diffusion processes. To parameterize this vector field, we employ a U-Net architecture, due to its encoder-decoder structure with skip connections, which allows it to effectively capture both high-level global structure (e.g. path connectivity) and low-level local details (e.g., tile adjacency). While the core algorithm remains consistent with Lipman et al.

(2023), the architecture will be adapted to handle discrete tile-based data through *one-hot encoding*, rather than continuous image data. To validate this approach, the proposed model will be empirically tested against a WFC baseline. The comparison will focus on quantitative metrics of sampling efficiency (NFE) and qualitative assessment of global coherence, aiming to demonstrate that Flow Matching can achieve the structural quality of diffusion models with the operational speed required for game development pipelines.

2.6 COMPARATIVE ANALYSIS FRAMEWORK

To synthesize the diverse approaches discussed in this chapter, we establish a comparative framework based on three critical criteria relevant to the goals of this thesis: Constraint Handling, Sampling Efficiency, and Data Requirements.

Constraint Handling evaluates the method’s ability to satisfy the game design rules defined in the introduction. This is a qualitative assessment based on the reported capabilities in the literature. An approach is rated as “Excellent” if it consistently produces playable maps that respect both local and global rules without extensive manual intervention. It is rated as “Poor” or “Limited” if it frequently produces invalid states, broken paths, or requires significant post-generation repairs (repairing algorithms).

Sampling Efficiency measures the computational cost and speed of generating a single valid map instance. For deep generative models, this is quantitatively estimated by the NFE. High Efficiency corresponds to methods that generate content in a single pass or very few steps (Low NFE, e.g. < 50), enabling near real-time generation. Low Efficiency corresponds to iterative methods requiring hundreds or thousands of steps (High NFE, e.g. > 500) to converge to a valid result, making them unsuitable for real-time applications.

Data Requirements distinguishes the input necessary to train or configure the algorithm. Single-Example methods like WFC typically require only a single input image or a small set of tiles to extract rules. Dataset-Driven methods like Deep learning approaches (GANs, DDPMs, Flow Matching) require a prepared dataset of valid maps. This involves pre-processing steps such as *chunking*—dividing large maps into smaller training tensors and *one-hot encoding* to represent categorical tile data, as described in the methodology of this work.

Approach	Representative Work	Constraint Handling	Sampling Efficiency	Data Requirement	Key Limitation
Standard WFC	Gumin (2016)	Excellent Local / Poor Global	Very High	Single Example	Rigid output; lack of global structure.
Controllable WFC	Cheng et al. (2021)	Improved Global via heuristics	Medium (Backtracking)	Single Example + Heuristics	Complexity of manual rule definition.
GANs	Silva et al. (2025)	Learns patterns; struggles with validity	High (Single pass)	Large Dataset	Mode collapse; training instability.
DDPMs	Ren (2024)	Excellent Global & Local	Low (High NFE)	Large Dataset	Very slow generation (1000s of steps).
Flow Matching	Lipman et al. (2023)	Excellent Global & Local	High (Low NFE)	Large Dataset	Emerging technique; requires ODE solver.

Table 2.1. Comparative analysis of generative approaches for procedural map generation

3 PROPOSED IMPLEMENTATION PLAN

Following the theoretical foundation and the comparative analysis of generative models, this chapter details the methodological approach designed to achieve the objectives of this study. The proposed solution consists of a complete pipeline for procedurally generating 2D game maps, treating the task as a continuous transformation from a noise distribution to a valid map distribution that satisfies implicit constraints.

The implementation is structured into three main stages: Data Engineering, Model Training (Conditional Flow Matching), and Inference (Euler-based Generation). The following sections describe the high-level architecture and the specific components of each stage.

3.1 SOLUTION ARCHITECTURE

The system is designed as a deep learning pipeline where a neural network approximates a time-dependent vector field. This vector field defines the optimal trajectory to transform random noise into a coherent game map. Figure 3.1 illustrates the high-level architecture of the proposed solution, detailing the flow of data from raw assets to the generated output.

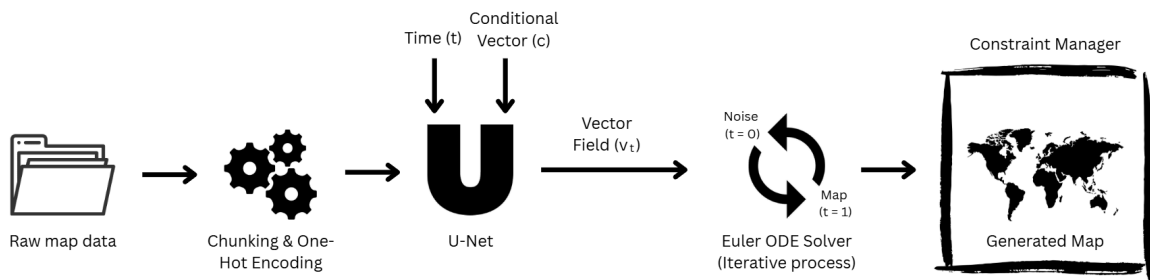


Figure 3.1: High-level architecture of the proposed Conditional Flow Matching generation pipeline.

As illustrated, the process begins with the extraction and preprocessing of tile-based maps into a tensor format suitable for the neural network. The core component is a U-Net architecture trained using the Conditional Flow Matching (CFM) objective. Finally, the inference

module utilizes an Euler ODE solver to generate new content, guided by conditional vectors and a constraint manager that ensures the output meets specific design requirements.

3.2 DATA ENGINEERING PIPELINE

To train the model to recognize valid local and global constraints (Objective O2), a robust dataset is required. The data pipeline consists of three steps:

1. **Data Acquisition:** The dataset will be constructed from 2D tile-based maps (e.g., extracted similar games or procedural generation benchmarks). These maps represent the "valid state" of the Constraint Satisfaction Problem (CSP), where the "energy" (constraint violations) is zero.
2. **Chunking:** Large game maps will be divided into smaller, fixed-size chunks (e.g., 32x32 or 64x64 tiles). This standardization is necessary for efficient batch processing during neural network training.
3. **One-Hot Encoding:** Unlike standard image generation which uses RGB channels, game maps are categorical. Each tile material (e.g., water, grass, sand) and geometry (e.g., square, rectangle, triangle) will be converted into one-hot encoded vectors. Consequently, the input to the model will be a tensor of shape (C, H, W), where C represents the number of unique tile types.

3.3 GENERATIVE MODEL: U-NET AND CONDITIONAL FLOW MATCHING

The core of the proposal is the substitution of the iterative diffusion process with a deterministic Flow Matching mechanism (Objective O3).

3.3.1 Network Architecture (U-Net)

We employ a U-Net architecture as the function approximator for the vector field. Originally designed for segmentation, the U-Net is ideal for this task due to its encoder-decoder structure with skip connections. While the encoder captures global context (global constraints), such as biome distribution and path connectivity, the decoder reconstructs fine details (local

constraints), ensuring correct tile adjacency. The network will receive the current time step t and a conditional vector c as additional inputs, injected into the network layers via adaptive normalization (AdaGN) or concatenation.

3.3.2 Training Objective (CFM)

The model is trained using the Conditional Flow Matching objective proposed by Lipman et al. (2023). Instead of simulating the full noise-to-data trajectory, the model learns to regress a target vector field $u_t(x|x_1)$ that points directly from the current noisy state towards the target data sample x_1 . This simulation-free training allows for faster convergence compared to traditional diffusion training.

3.4 INFERENCE AND CONSTRAINT MANAGEMENT

The generation process is framed not just as image synthesis, but as an iterative planning process to minimize constraint violations, which will be addressed as “energy” in this work.

The inference process starts with random noise (high entropy/high violation state). The Euler solver iteratively applies the vector field to "move" the map configuration towards a valid state (zero violations). This parallels the concept of annealing, where the system gradually settles into a low-energy configuration.

To ensure the generated maps meet specific design requirements, the generation process is guided by a conditional vector c . This vector encodes high-level attributes (e.g., "Desert Biome", "Contains Lava").

Constraint Manager (Proposed Component): To explicitly validate the generated maps, a "Constraint Manager" module will be implemented. This component will firstly take the generated board as input, then perform a reverse lookup to identify tile rules, and finally, calculate the total number of constraint violations (local adjacency errors, global path failures). This manager provides the quantitative metric for evaluating the model's success in solving the Constraint Satisfaction Problem.

3.5 VALIDATION STRATEGY

The implemented solution will be evaluated based on the comparative framework established in Chapter 2. The primary metrics for validation will be:

1. Sampling Efficiency: Measured by the NFE required to produce a valid map and the total wall-clock time for generation.
2. Constraint Satisfaction: Evaluated by the Constraint Manager, calculating the percentage of valid tile adjacencies (local) and the structural coherence of biomes (global).
3. Diversity: Assessed to ensure the model does not suffer from mode collapse, comparing it against the output of a Wave Function Collapse (WFC) or Genetic Algorithm baseline.

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